Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

# Homework Session 3

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BUSINESS DATA MINING

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UNIVERSITY OF DUBLIN

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## Appendix

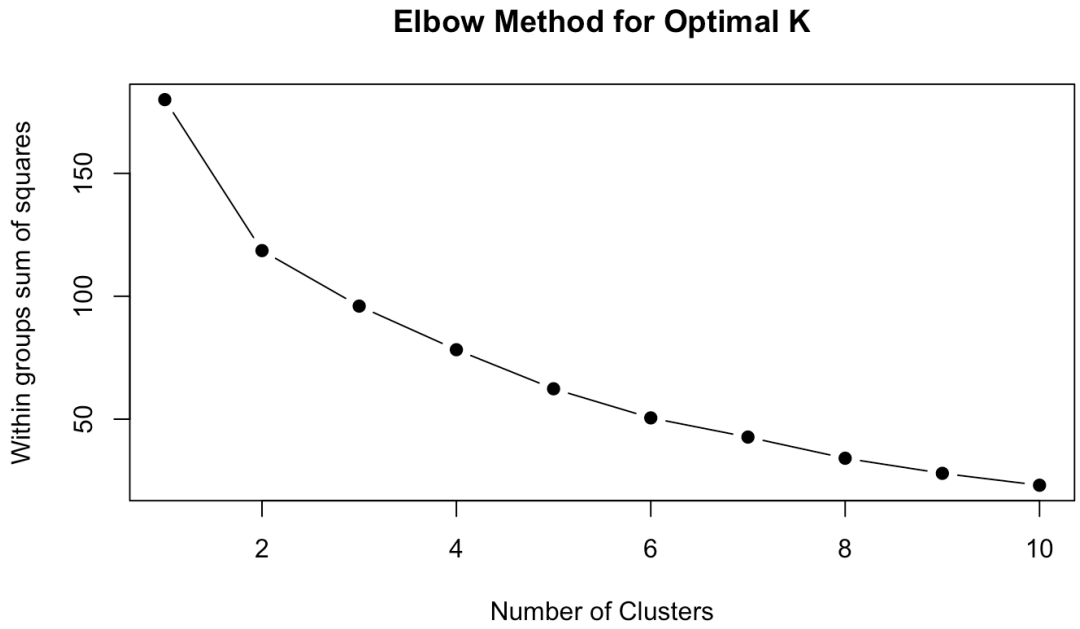
I used DeepSeek, copilot. I used copilot when I used R to code, which helps to check the errors. I used DeepSeek on code, which gave me ideas, and on translations, because English is not my first language. For example, I would ask DeepSeek what the statement means.

Credit is given to Christianto Aryanto (24345328) for assistance in a discussion about the hierarchical problem in East West Airlines case.

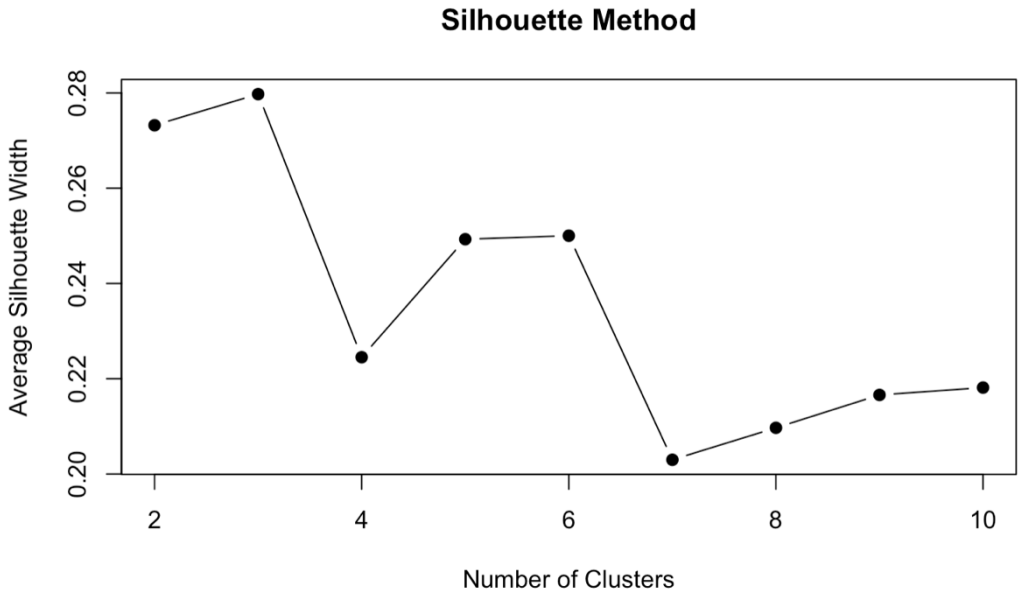
# Business Data Mining Homework Session 3

Q15.2 Pharmaceutical Industry. An equities analyst is studying the pharmaceutical industry and would like your help in exploring and understanding the financial data collected by her firm. Her main objective is to understand the structure of the pharmaceutical industry using some basic financial measures.

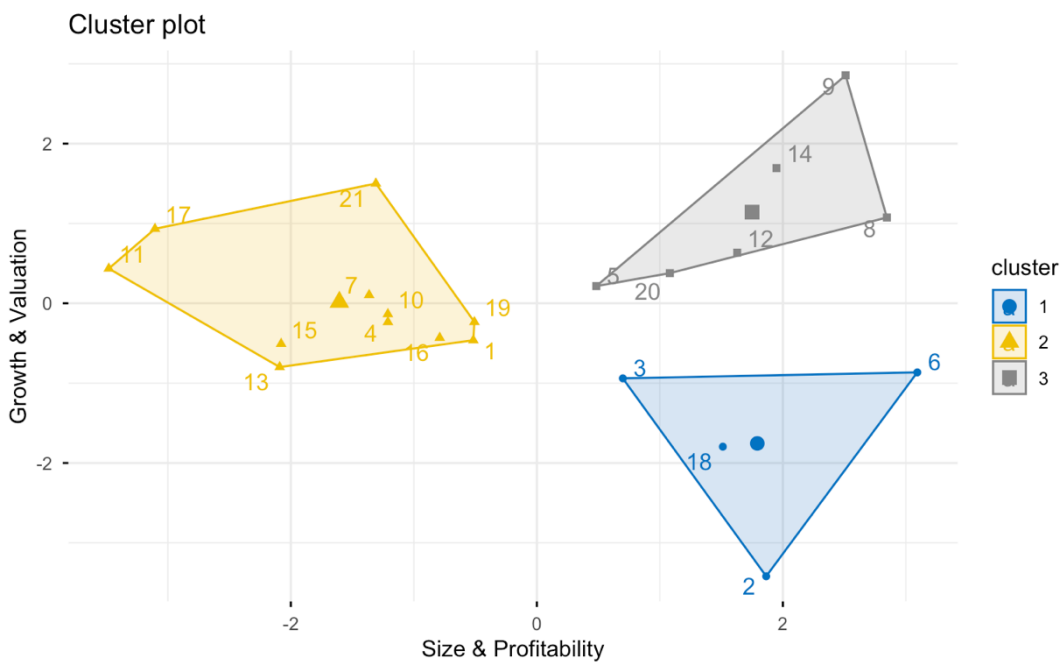
a. Use only the numerical variables (1 to 9) to cluster the 21 firms. Justify the various choices made in conducting the cluster analysis, such as weights for different variables, the specific clustering algorithm(s) used, the number of clusters formed, and so on.

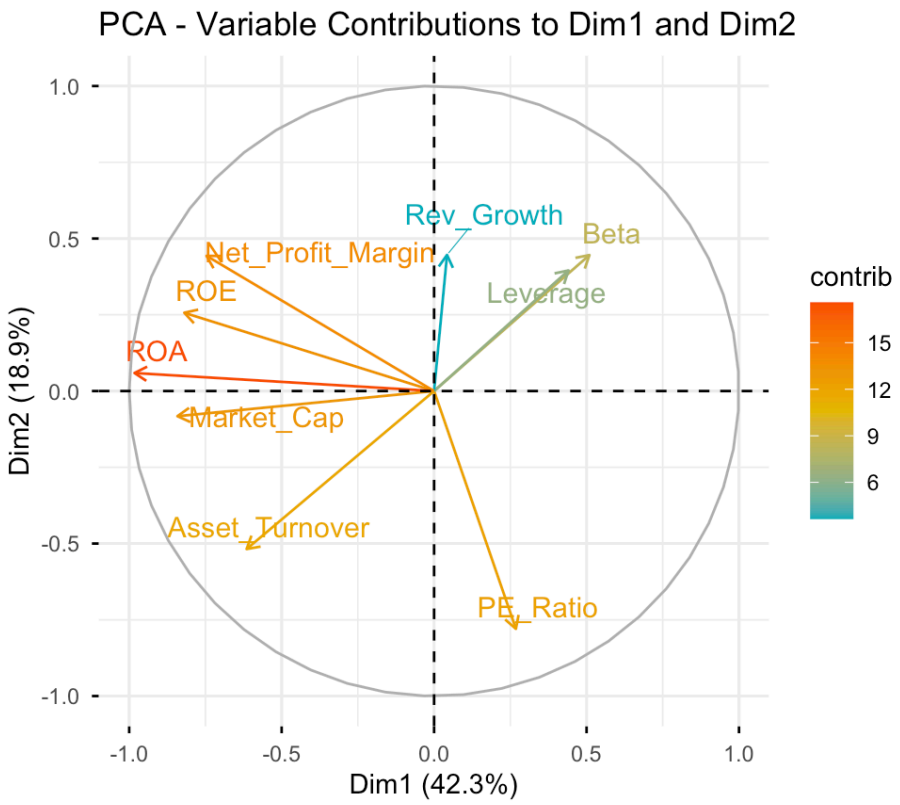



The elbow method suggests k=2 because WSS improvement reduces after this point. Although the inflection is not sharply defined. Given we're analysing 21 companies, 2 clusters may be too simplistic to capture meaningful business segments. Therefore, I employed additional method to determine an optimal k and to ensures statistical robustness and practical business relevance for identifying distinct customer segments.



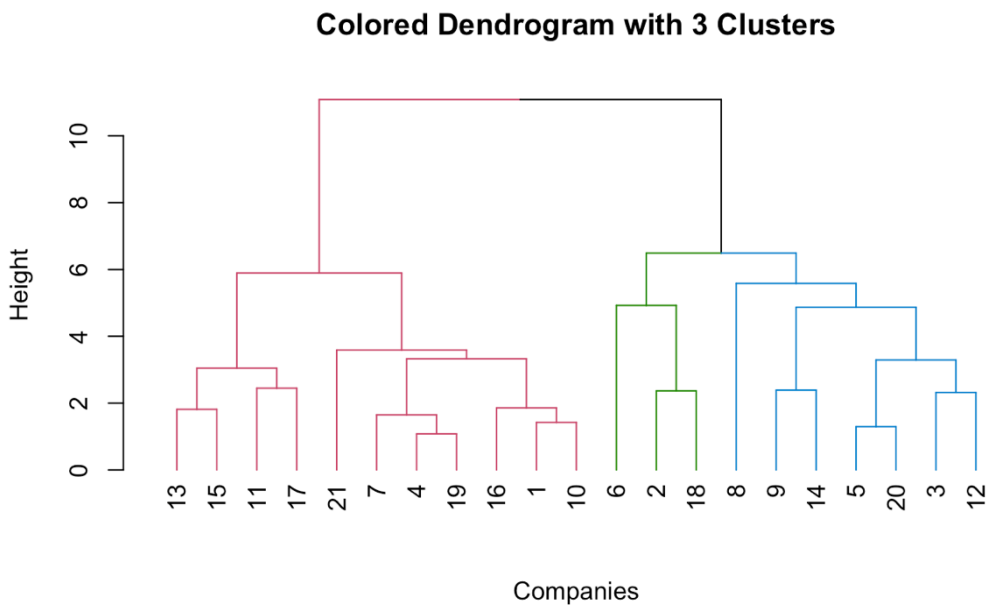

I used silhouette method and find the highest width is 0.28, where k = 3. According to interpretation guidelines, a score of 0.28 indicates a weak and unstructured partition. This suggests that 21 companies exhibit limited natural grouping patterns and may not create clearly distinct segments based on the available variables.






To find out the dimension1 and dimension2, I use PCA method. Here we can find that dimension1 is primarily defined by Market Cap, ROA, ROE, and Net Profit Margin (all with strong negative loadings). This represents a "Size and Profitability" dimension, where companies scoring high on dimension1 have smaller market caps and lower profitability metrics. Dimension 2 is characterized by high Revenue Growth and Beta (positive loadings) but low PE Ratio (strong negative loading). This reveals a "Growth and Valuation" dimension, where companies with high growth are perceived as riskier, leading the market to assign them lower valuations due to reduced confidence in their prospects.

For comparative purposes, hierarchical clustering was employed.



The cluster assignments remain largely consistent between methods, with the exception of Company 3, which shifted from the green group. This overall stability validates the robustness of the identified cluster structure.

b. Interpret the clusters with respect to the numerical variables used in forming the clusters.

Cluster 1 (Positive on Size & Profitability, Negative on Growth & Valuation)
This cluster represents large, profitable, but low-growth companies. These established firms generate strong profits but face market scepticism about future growth, resulting in lower market values. They are typically mature, dividend-paying value stocks.
Cluster 2 (Negative on Size & Profitability, near-zero on Growth & Valuation)
This cluster contains smaller, less profitable companies with stagnant growth. They lack both the scale of Cluster 1 and the growth story of Cluster 3, positioning them as potential "value traps" or firms in transition.
Cluster 3 (Positive on both axes)
This is the "High-Quality Growth" cluster. These companies combine strong profitability with high growth. Market valuations don't fully reflect this potential, suggesting they may be undervalued and have growth opportunities.

c. Is there a pattern in the clusters with respect to the nun-numerical variables (10 to 12)?

Location Distribution:

| CANADA | FRANCE | GERMANY | IRELAND | SWITZERLAND | UK | US |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 1 | 2 | 8 |
| 0 | 1 | 0 | 1 | 0 | 0 | 4 |

Exchange Distribution:

| AMEX | NASDAQ | NYSE |
|---|---|---|
| 0 | 0 | 4 |
| 0 | 0 | 11 |
| 1 | 1 | 4 |

Analyst Recommendation Distribution:

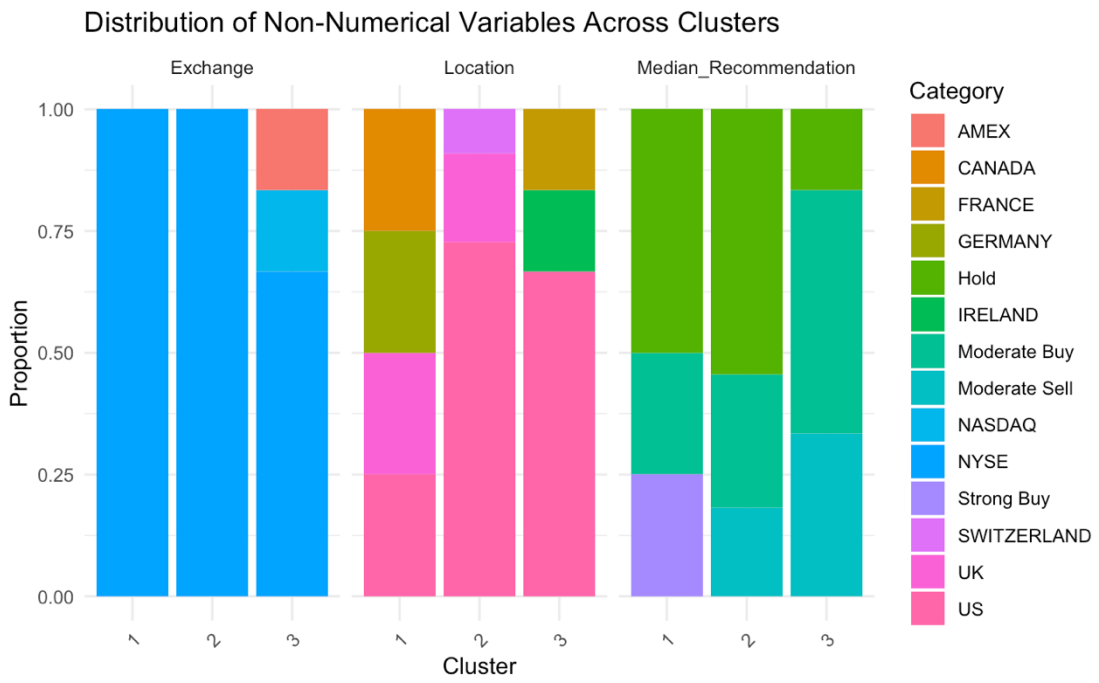
| Hold | Moderate Buy | Moderate Sell | Strong Buy |
|---|---|---|---|
| 2 | 1 | 0 | 1 |
| 6 | 3 | 2 | 0 |
| 1 | 3 | 2 | 0 |

Location Percentages:

| CANADA | FRANCE | GERMANY | IRELAND | SWITZERLAND | UK | US |
|---|---|---|---|---|---|---|
| 25 | 0 | 25 | 0 | 0 | 25 | 25 |
| 0 | 0 | 0 | 0 | 9.1 | 18.2 | 72.7 |
| 0 | 16.7 | 0 | 16.7 | 0 | 0 | 66.7 |

Recommendation Percentages:

| Hold | Moderate Buy | Moderate Sell | Strong Buy |
|---|---|---|---|
| 50 | 25 | 0 | 25 |
| 54.5 | 27.3 | 18.2 | 0 |
| 16.7 | 50 | 33.3 | 0 |




Cluster 1 "Diversified Special Niche"
The most diversified cluster, with companies located in Canada, Germany, the UK, and the US, with no single country dominating (each at 25%). This suggests it is not defined by geography but potentially by specific business models or niche markets. 100% listed on the New York Stock Exchange (NYSE), indicating these are larger, more established companies, fitting the NYSE profile. It is the only cluster to receive a "Strong Buy" rating (25%), while also having a mix of "Hold" and "Moderate Buy". This indicates strong market optimism for some firms within it, but overall opinion is divided.

Cluster 2 "US-Dominated Defensive"
Highly concentrated, with 72.7% of firms headquartered in the US, and the rest in Switzerland and the UK. This is a strongly American-centric cluster. 100% listed on the New York Stock Exchange (NYSE), reinforcing its character as dominated by large US corporations. The most negatively viewed cluster. Over half (54.5%) received a "Hold" rating, and 18.2% received a "Moderate Sell" rating, with no "Strong Buy" ratings. This perfectly aligns with the market's expectations for "value/defensive" or "distressed" companies seen as having limited growth potential and lacking analyst enthusiasm.

Cluster 3 "Euro-American Growth"
Led by the US (66.7%), but mixed with European presence (France and Ireland at 16.7% each). This is an international growth-oriented cluster. A unique and mixed pattern. It is the only cluster containing listings outside the NYSE (1 on AMEX, 1 on NASDAQ). This reflects the diversity and potentially different growth stages of its constituent companies. The most divisive cluster for market opinion. While it has no "Strong Buy" ratings, it has the highest proportion of "Moderate Buy" ratings (50%). However, it also has

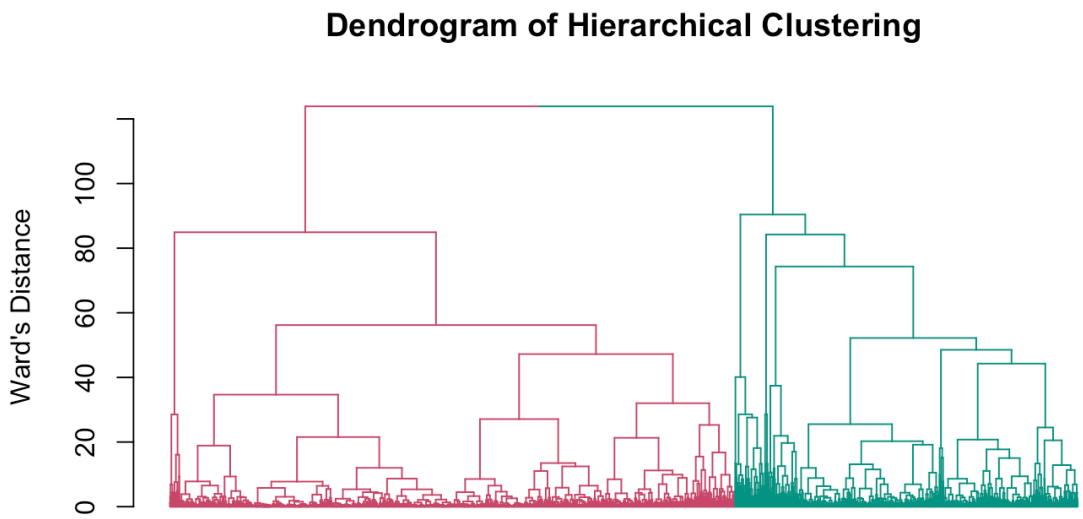
the highest "Moderate Sell" rating proportion (33.3%), indicating that market sentiment is the most polarized for this cluster, showing significant disagreement.

d. Provide an appropriate name for each cluster using any or all of the variables in the dataset.

Cluster 1: "Stable Quality Giants"
Cluster 2: "Value Traps"
Cluster 3: "Controversial Growth Stories"

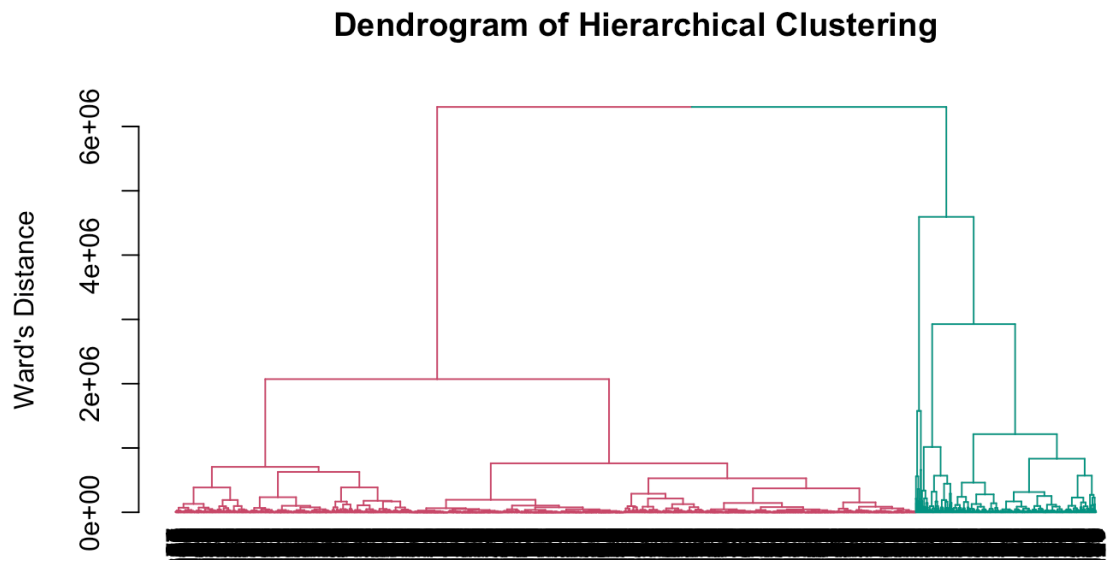
Q15.4 Marketing to Frequent Fliers. The file *EastWestAirlinesCluster.csv* contains information on 3999 passengers who belong to an airline's frequent flier program. For each passenger, the data include information on their mileage history and on different ways they accrued or spent miles in the last year. The goal is to try to identify clusters of passengers that have similar characteristics for the purpose of targeting different segments for different types of mileage offers.

a. Apply hierarchical clustering with Euclidean distance and Ward's method. Make sure to normalize the data first. How many clusters appear?




The dendrogram reveals 2 distinct clusters. This is evident from the two primary branches separated by the largest vertical distance on the diagram.

b. What would happen if the data were not normalized?



From the diagram, there is not much difference. However, it is obvious that the vertical coordinate units are different, and the red cluster has a larger range on the vertical coordinate. Variables with large values will dominate the distance calculation, and the influence of variables with small values will be ignored. Clustering is mainly based on high-value variables, and low-value but business-significant variables are ignored, which may result in clusters without business significance. I think we should normalize data. Otherwise, the clustering results may not be meaningful to the business and may only reflect the magnitude differences in the data rather than the actual customer behaviour patterns.

c. Compare the cluster centroid to characterize the different clusters, and try to give each cluster a label.

| cluster | Balance | Qual_miles | cc1_miles | cc2_miles | cc3_miles |
|---|---|---|---|---|---|
| 1 | -0.267 | -0.174 | -0.594 | 0.0596 | -0.0628 |
| 2 | 0.440 | 0.287 | 0.978 | -0.0982 | 0.1034 |

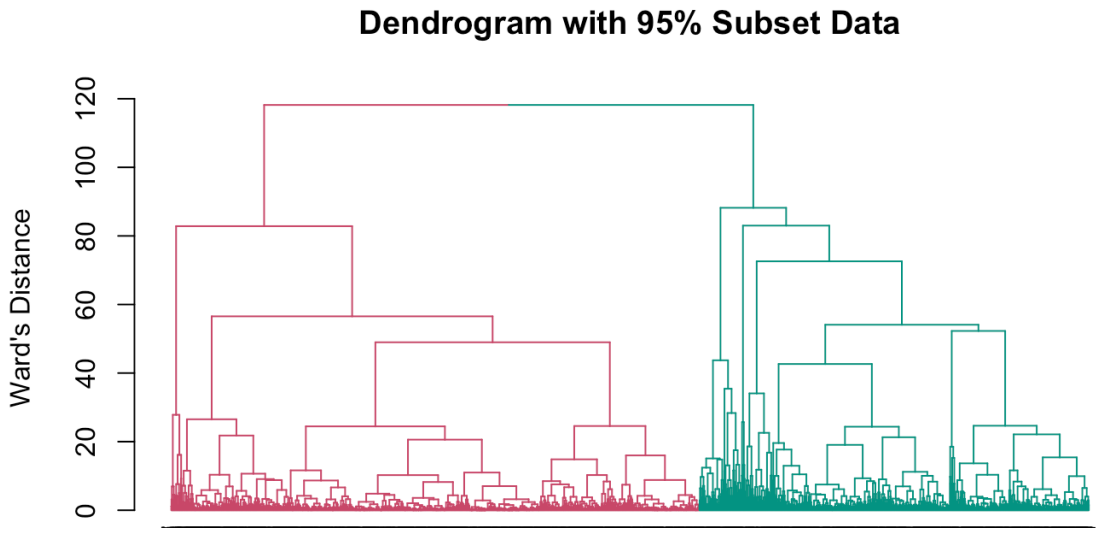
| Bonus_miles | Bonus_trans | Flight_miles_12mo | Flight_trans_12 | Days_since_enroll | Award |
|---|---|---|---|---|---|
| -0.501 | -0.470 | -0.171 | -0.178 | -0.167 | -0.377 |
| 0.826 | 0.774 | 0.281 | 0.293 | 0.276 | 0.622 |

Variables can be categorized into four key business dimensions for customer segmentation analysis.
Spending Capacity Dimension includes Balance, Bonus_miles. They reflect customers' overall mileage accumulation and purchasing power.
Flight Activity Dimension includes Flight_miles_12mo, Flight_trans_12, and Qual_miles. They measure recent flight frequency, distance traveled, and eligibility mileage.
Member Loyalty Dimension includes cc1_miles, cc2_miles, cc3_miles, and Days_since_enroll. They indicate membership tier status and customer relationship duration.
Promotion Responsiveness Dimension includes Bonus_trans and Award. They capture engagement withpromotional offers and reward redemption behavior.

Cluster 1: Development Passengers
This group shows below-average activity across all metrics, such as low mileage balance, minimal bonus transactions, and limited flight activity. They are newer members with low credit card usage, representing occasional travelers with significant growth potential through targeted engagement campaigns.

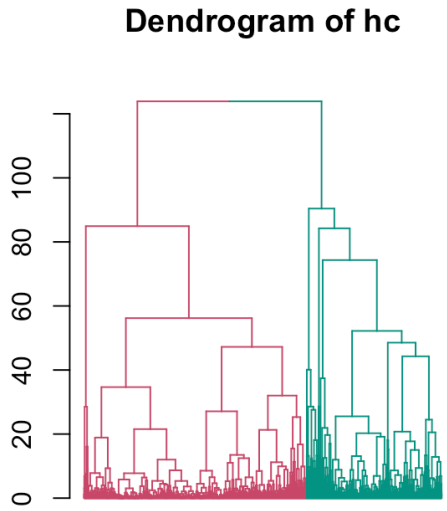
Cluster 2: High-Value Frequent Fliers
These passengers demonstrate above-average performance in all areas, such as high mileage balances, frequent bonus transactions, and substantial flight activity. They heavily utilize premium credit cards and represent the airline's core profitable segment, ideal for premium loyalty rewards and business-class promotions.

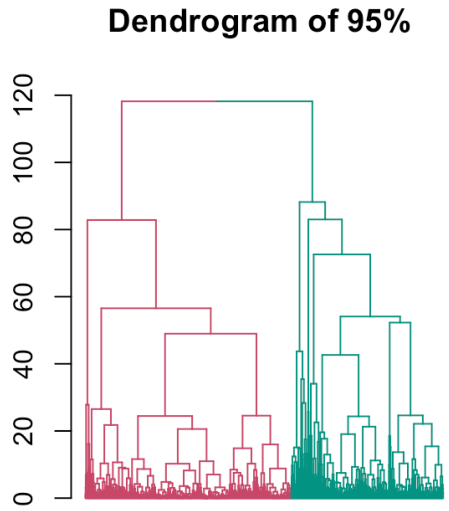
d. To check the stability of the clusters, remove a random 5% of the data (by taking a random sample of 95% of the records), and repeat the analysis. Does the same picture emerge?




| cluster | Balance | Qual_miles | cc1_miles | cc2_miles | cc3_miles |
|---|---|---|---|---|---|
| 1 | -0.2694919 | -0.1676751 | -0.6806748 | 0.07438325 | -0.06132063 |
| 2 | 0.3652259 | 0.2272398 | 0.9224769 | -0.10080706 | 0.08310409 |

| Bonus_miles | Bonus_trans | Flight_miles_12mo | Flight_trans_12 | Days_since_enroll | Award |
|---|---|---|---|---|---|
| -0.5399520 | -0.5200683 | -0.2097338 | -0.2194872 | -0.1363151 | -0.3324516 |
| 0.7317639 | 0.7048166 | 0.2842393 | 0.2974576 | 0.1847395 | 0.4505512 |



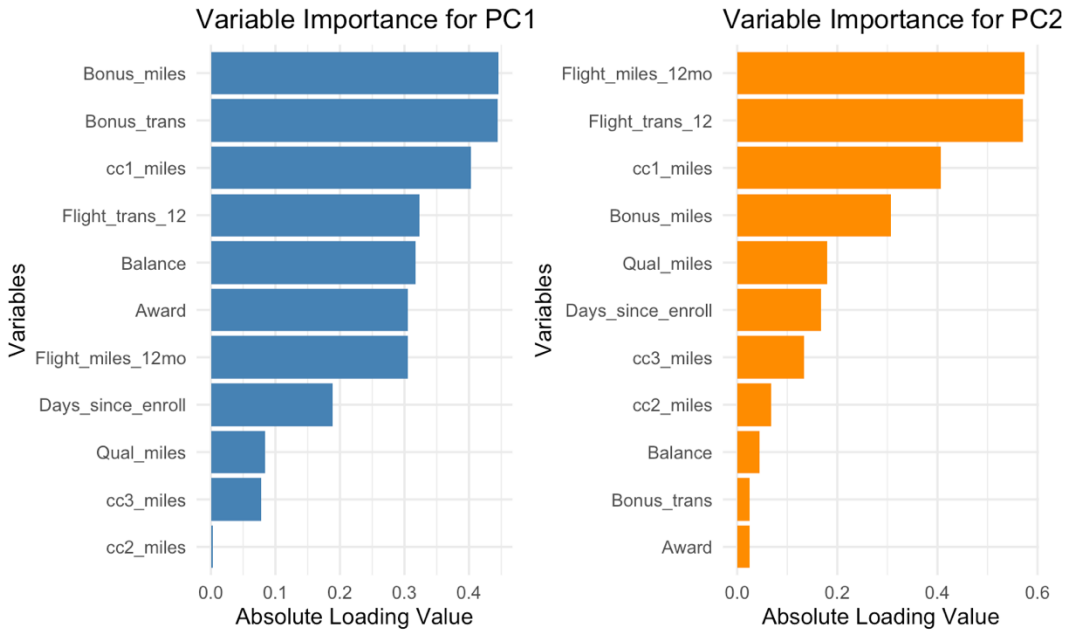

The stability assessment confirms robust clustering results. The centroid values and dendrogram remain virtually after removing a random 5% of the data. This consistency demonstrates that the identified two-cluster solution is not sensitive to minor data variations and represents a stable segmentation of the airline passenger base.

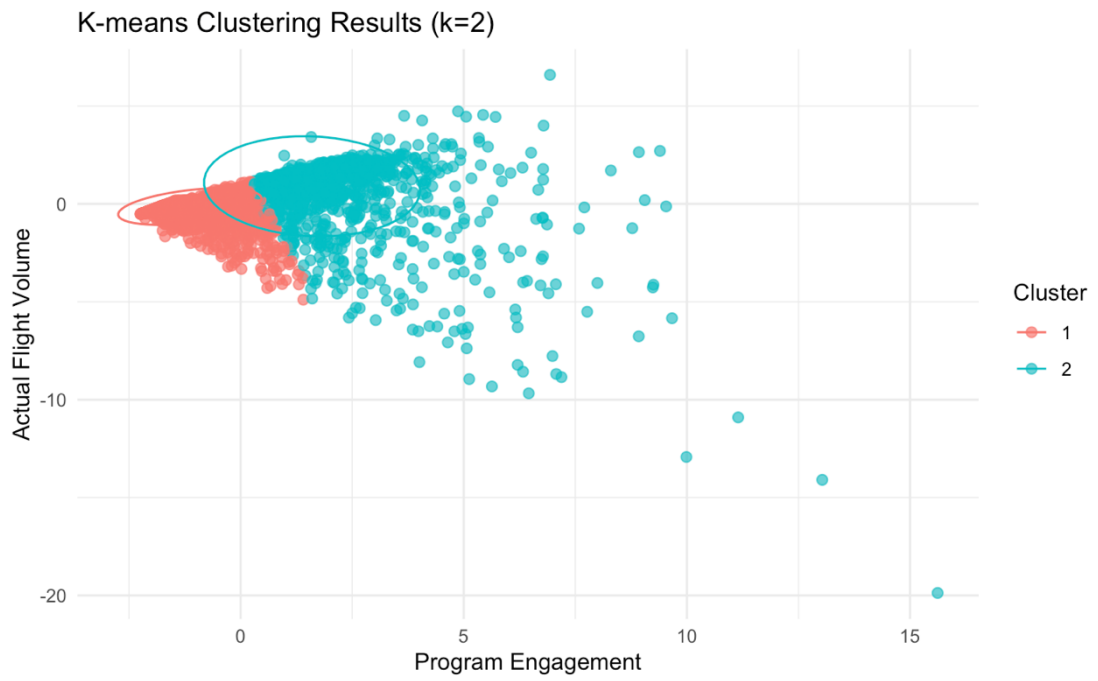
e. Use k-means clustering with the number of clusters that you found above. Does the same picture emerge?

| cluster | Balance | Qual_miles | cc1_miles | cc2_miles | cc3_miles |
|---|---|---|---|---|---|
| 1 | -0.283 | -0.0711 | -0.542 | 0.0146 | -0.0609 |
| 2 | 0.588 | 0.1477 | 1.128 | -0.0304 | 0.1265 |

| Bonus_miles | Bonus_trans | Flight_miles_12mo | Flight_trans_12 | Days_since_enroll | Award |
|---|---|---|---|---|---|
| -0.485 | -0.440 | -0.175 | -0.195 | -0.192 | -0.338 |
| 1.008 | 0.915 | 0.364 | 0.405 | 0.399 | 0.703 |

Applying k-means clustering with k=2 highly consistent results compared to the hierarchical method. The cluster centroids from both methods show remarkable alignment, revealing the same fundamental customer segmentation pattern. This strengthens the validity of the cluster solution for profiling airline passengers.






The resulting clusters clearly align with the principal components identified through PCA. Cluster 1 corresponds to customers with high engagement in bonus activities (aligned with PC1), while Cluster 2 represents frequent flyers with substantial flight mileage and transactions (aligned with PC2). This convergence between PCA dimensions and k-means clusters confirms that the underlying customer segments are robust and methodologically consistent.

f. Which clusters would you target for offers, and what types of offers would you target to customers in that cluster?

Cluster 1 shows low-value or occasional travellers. They have characteristics, such as below-average activity across all metrics, newer members, limited engagement.

Cluster 2 shows high-value or frequent business travellers. They have the characteristics, such as high mileage balances, frequent transactions, premium credit card users.

I will implement both strategies simultaneously but with different messaging and channel approaches tailored to each cluster's behaviour patterns. For instance, for travellers in Cluster 1, I would offer a 30% discount on selected regional routes or a membership tier upgrade upon completing three flights within three months, and regularly push promotional information via email. For travellers in Cluster 2, I would provide complimentary lounge access passes or complimentary room upgrades at partner hotels, delivering these offers through dedicated account managers for personalized service.